

Week - 3

2520030030

U.Rohit Sai

1. Create a database bookflow

The screenshot shows the MongoDB Atlas Data Explorer interface. On the left, the 'Data Explorer' sidebar displays a tree view of clusters. Under 'Cluster0', the 'bankdb' database is expanded, showing collections 'customers' and 'transactions'. The main panel shows the 'bankdb' database selected, with a table listing the collections. The table has columns: Collection name, Properties, Storage size, Data size, Documents, Avg. document size, Indexes, and Total index size.

Collection name	Properties	Storage size	Data size	Documents	Avg. document size	Indexes	Total index size
customers	-	36.86 kB	1.11 kB	8	138.00 B	1	36.86 kB
transactions	-	36.86 kB	2.19 kB	15	145.00 B	1	36.86 kB

2. Create a new collection book_metadata

The screenshot shows the MongoDB Atlas Data Explorer interface. The 'Data Explorer' sidebar on the left shows the 'bookflow' database expanded, with the 'book_metadata' collection highlighted. The main panel shows the 'bookflow' database selected, with a table listing the collections. The table has columns: Collection name, Properties, Storage size, Data size, Documents, Avg. document size, Indexes, and Total index size.

Collection name	Properties	Storage size	Data size	Documents	Avg. document size	Indexes	Total index size
book_metadata	-	36.86 kB	2.19 kB	15	145.00 B	1	36.86 kB

3. Click add data and insert documents

The screenshot shows the MongoDB Atlas Data Explorer interface. The 'Data Explorer' sidebar on the left shows the 'bookflow' database expanded, with the 'book_metadata' collection highlighted. The main panel shows the 'bookflow' database selected, with a table listing the collections. The table has columns: Collection name, Properties, Storage size, Data size, Documents, Avg. document size, Indexes, and Total index size.

Collection name	Properties	Storage size	Data size	Documents	Avg. document size	Indexes	Total index size
book_metadata	-	36.86 kB	2.19 kB	15	145.00 B	1	36.86 kB

The 'Documents' tab is selected, showing a list of 5 documents. The documents are:

Document ID	book_id	title	author	published_year
1	101	"The Martian"	"Andy Weir"	2021
2	102	"Dune"	"Frank Herbert"	2022
3	103	"Clean Code"	"Robert C. Martin"	2019
4	104	"Artificial Intelligence ..."	"John Smith"	2023
5	105	"Data Structures"	"Mark Allen"	2024

4. Mongo DB Queries :

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Data Explorer

My Queries Data Modeling

CLUSTERS (1)

Search clusters

- Cluster0
 - admin
 - bankdb
 - customers
 - transactions
 - bookflow
 - book_metadata
 - local
 - sample_mflix

book_metadata +

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

Generate query + Explain Reset Find Options

```
{
  "reviews.rating": {
    "$gte": 4
  }
}
```

ADD DATA BULK EXPORT CODE 25 1-5 of 5

	_id ObjectId	book_id Int32	title String	author String	published_year Int32	
1	ObjectId('6a7eb69b6d946a9...')	101	"The Martian"	"Andy Weir"	2021	
2	ObjectId('6a7eb74db16ff8f...')	102	"Dune"	"Frank Herbert"	2022	
3	ObjectId('6a7eb768b16ff8f...')	103	"Clean Code"	"Robert C. Martin"	2019	
4	ObjectId('6a7eb772b16ff8f...')	104	"Artificial Intelligence ..."	"John Smith"	2023	
5	ObjectId('6a7eb78fb16ff8f...')	105	"Data Structures"	"Mark Allen"	2024	

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Data Explorer

My Queries Data Modeling

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book_metadata +

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

Generate query + Explain Reset Find Options

```
{
  "reviews.rating": {
    "$in": [4, 5]
  }
}
```

ADD DATA BULK EXPORT CODE 25 1-5 of 5

	_id ObjectId	book_id Int32	title String	author String	published_year Int32	
1	ObjectId('6a7eb69b6d946a9...')	101	"The Martian"	"Andy Weir"	2021	
2	ObjectId('6a7eb74db16ff8f...')	102	"Dune"	"Frank Herbert"	2022	
3	ObjectId('6a7eb768b16ff8f...')	103	"Clean Code"	"Robert C. Martin"	2019	
4	ObjectId('6a7eb772b16ff8f...')	104	"Artificial Intelligence ..."	"John Smith"	2023	
5	ObjectId('6a7eb78fb16ff8f...')	105	"Data Structures"	"Mark Allen"	2024	

5. Aggregations

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Data Explorer

My Queries Data Modeling

CLUSTERS (1)

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book_metadata +

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

Generate aggregation + Explain Run Options

Untitled - modified SAVE CREATE NEW EXPORT CODE PREVIEW STAGES TEXT WIZARD

Stage 1 \$match

```
{
  "published_year": {
    "$gt": 2020
  }
}
```

Output preview after \$match stage (Sample of 4 documents)

```
{
  "_id": "ObjectId('6a7eb69b6d946a9b6e6a7954b')",
  "book_id": 101,
  "title": "The Martian",
  "author": "Andy Weir",
  "published_year": 2021,
  "category": "Science Fiction",
  "format": "Digital PDF",
  "file_size": "2.5 MB",
  "member_id": 101
}
```

```
{
  "_id": "ObjectId('6a7eb74db16ff8f2')",
  "book_id": 102,
  "title": "Dune",
  "author": "Frank Herbert",
  "published_year": 2022,
  "category": "Science Fiction",
  "format": "Digital EPUB",
  "file_size": "3.1 MB",
  "member_id": 102
}
```

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Data Explorer

🔍 My Queries

🏗️ Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- bankdb
- bookflow
- book_metadata**
- local
- sample_mflix

book_metadata

+

Cluster0 > bookflow > book_metadata

📄 Documents 5

Aggregations

Schema

Indexes 1

Validation

📄 View monitoring

📊 Visualize Your Data

🔍

📄 \$match

Sunwind

Generate aggregation

🔍 Explain

Run

Options

Untitled - modified

📄 SAVE

+ CREATE NEW

📄 EXPORT CODE

PREVIEW

STAGES

📄 TEXT

WIZARD

⚙️

Stage 2

\$unwind

🔍

📄

📄

1 "\$reviews"

📄

📄

Output preview after \$unwind stage (Sample of 10 documents)

```
_id: ObjectId('6a7eb69bd946a9be6a7954b')
book_id: 101
title: "The Martian"
author: "Andy Weir"
published_year: 2021
category: "Science Fiction"
format: "Digital PDF"
file_size: "2.5 MB"
member_id: 101
```

```
_id: ObjectId('6a7eb69bd946a9be6a7954b')
book_id: 101
title: "The Martian"
author: "Andy Weir"
published_year: 2021
category: "Science Fiction"
format: "Digital PDF"
file_size: "2.5 MB"
member_id: 101
```

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Data Explorer

🔍 My Queries

🏗️ Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- bankdb
- bookflow
- book_metadata**
- local
- sample_mflix

book_metadata

+

Cluster0 > bookflow > book_metadata

📄 Documents 5

Aggregations

Schema

Indexes 1

Validation

📄 View monitoring

📊 Visualize Your Data

🔍

📄 \$match

Sunwind

\$group

Generate aggregation

🔍 Explain

Run

Options

Untitled - modified

📄 SAVE

+ CREATE NEW

📄 EXPORT CODE

PREVIEW

STAGES

📄 TEXT

WIZARD

⚙️

Stage 3

\$group

🔍

📄

📄

1 {

2 _id: "\$title",

3 average_rating: {

4 \$avg: "\$reviews.rating"

5 },

6 total_reviews: {

7 \$sum: 1

8 }

9 }

📄

📄

Output preview after \$group stage (Sample of 4 documents)

```
_id: "The Martian"
average_rating: 4.666666666666667
total_reviews: 3
```

```
_id: "Dune"
average_rating: 4.666666666666667
total_reviews: 3
```

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Data Explorer

🔍 My Queries

🏗️ Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- bankdb
- bookflow
- book_metadata**
- local
- sample_mflix

book_metadata

+

Cluster0 > bookflow > book_metadata

📄 Documents 5

Aggregations

Schema

Indexes 1

Validation

📄 View monitoring

📊 Visualize Your Data

🔍

📄 \$match

Sunwind

\$group

\$sort

Generate aggregation

🔍 Explain

Run

Options

Untitled - modified

📄 SAVE

+ CREATE NEW

📄 EXPORT CODE

PREVIEW

STAGES

📄 TEXT

WIZARD

⚙️

Stage 4

\$sort

🔍

📄

📄

1 {

2 average_rating: -1

3 }

📄

📄

Output preview after \$sort stage (Sample of 4 documents)

```
_id: "Dune"
average_rating: 4.666666666666667
total_reviews: 3
```

```
_id: "The Martian"
average_rating: 4.666666666666667
total_reviews: 3
```

6. Indexes

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Data Explorer

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- bankdb
- bookflow
 - book_metadata**
 - local
 - sample_mflix

book_metadata

+

Cluster0 > bookflow > book_metadata

View monitoring Visualize Your Data

Documents 5 Aggregations Schema **Indexes 2** Validation

Create Index Refresh

Looking for search indexes?

These indexes can be created and viewed under [Search and Vector Search](#).

Name & Definition	Type	Size	Usage	Properties	Status
> _id_	REGULAR ⓘ	36.9 kB	3 (since Tue Aug 25 2026)	UNIQUE ⓘ	READY
> reviews.comments_text	TEXT ⓘ	20.5 kB	0 (since Thu Aug 27 2026)		READY

7. Search using text

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Data Explorer

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- bankdb
- bookflow
 - book_metadata**
 - local
 - sample_mflix

book_metadata

bookflow

+

Cluster0 > bookflow > book_metadata

View monitoring Visualize Your Data

Documents 5 Aggregations Schema Indexes 2 Validation

Generate query ⓘ Explain Reset Find Options

```
{  "$text": {    "$search": "excellent"  }}
```

ADD DATA BULK EXPORT CODE

25 1 - 5 of 5

book_metadata

_id	ObjectId	book_id	Int32	title	String	author	String	published_year	Int32
1	ObjectId('6a7eb69b6d946a9...')	101		"The Martian"		"Andy Weir"		2021	
2	ObjectId('6a7eb768b16ff8f...')	103		"Clean Code"		"Robert C. Martin"		2019	
3	ObjectId('6a7eb74db16ff8f...')	102		"Dune"		"Frank Herbert"		2022	
4	ObjectId('6a7eb78fb16ff8f...')	105		"Data Structures"		"Mark Allen"		2024	
5	ObjectId('6a7eb772b16ff8f...')	104		"Artificial Intelligence ..."		"John Smith"		2023	

8. Search for a specific word

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Data Explorer

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- bankdb
- bookflow
 - book_metadata**
 - local
 - sample_mflix

book_metadata

bookflow

+

Cluster0 > bookflow > book_metadata

View monitoring Visualize Your Data

Documents 5 Aggregations Schema Indexes 2 Validation

Generate query ⓘ Explain Reset Find Options

```
{  "$text": {    "$search": "\"science fiction\""  }}
```

ADD DATA BULK EXPORT CODE

25 1 - 2 of 2

book_metadata

_id	ObjectId	book_id	Int32	title	String	author	String	published_year	Int32
1	ObjectId('6a7eb74db16ff8f...')	102		"Dune"		"Frank Herbert"		2022	
2	ObjectId('6a7eb69b6d946a9...')	101		"The Martian"		"Andy Weir"		2021	